Robot Hardware Map

Every device, the subsystem that owns it, and where it plugs in. **Fill the blanks in pencil** as you wire: circle **CH** (Control Hub) or **EH** (Expansion Hub) and write the **Port #**. The **config name** must match the Driver Station config **exactly** (capital letters too). Then copy the ports into `hardware-config.md`.

Subsystem	Config name (Driver Station)	Device	Config type	Hub (circle)	Port type	Port #	Notes
Drivetrain	**`frontLeftDrive`**	goBILDA 5203 Yellow Jacket	goBILDA 5203 Series	CH / EH	Motor		Direction REVERSE
Drivetrain	**`backLeftDrive`**	goBILDA 5203 Yellow Jacket	goBILDA 5203 Series	CH / EH	Motor		Direction REVERSE
Drivetrain	**`frontRightDrive`**	goBILDA 5203 Yellow Jacket	goBILDA 5203 Series	CH / EH	Motor		
Drivetrain	**`backRightDrive`**	goBILDA 5203 Yellow Jacket	goBILDA 5203 Series	CH / EH	Motor		
Odometry	**`pinpoint`**	goBILDA Pinpoint + 2 pods	goBILDA Pinpoint	CH / EH	I2C bus		Not bus 0 (built-in IMU)
Odometry	**`imu`**	Control Hub built-in IMU	(built in)	**CH**	I2C bus	**0**	Built in, can't move
Intake	**`intakeServo`**	Servo (continuous)	Continuous Rotation Servo	CH / EH	Servo		Pulls game elements in
Intake	**`intakeLoaderServo`**	Servo (continuous)	Continuous Rotation Servo	CH / EH	Servo		Moves them up to the launcher
Intake	**`intake_____Switch`**	goBILDA micro limit switch	Digital Device	CH / EH	Digital		Job + name TBD (#4)
Intake	**`intake_____Switch`**	goBILDA micro limit switch	Digital Device	CH / EH	Digital		Job + name TBD (#4)
Launcher	**`launcherMotor`**	goBILDA 5203 Yellow Jacket	goBILDA 5203 Series	CH / EH	Motor		The flywheel
Launcher	**`launcherTriggerServo`**	Servo	Servo	CH / EH	Servo		REST / FIRE
Turret	**`turretRotateServo`**	goBILDA 2000 25-3 Speed (continuous)	Continuous Rotation Servo	CH / EH	Servo		Signal 1000-2000 us
Turret	**`turretEncoder`**	ELC Encoder V2, "D" cable	Motor (never powered)	**EH**	Encoder		Motor port # must match; keep that motor port empty
Turret	**`turretForwardSwitch`**	REV Magnetic Limit Switch	Digital Device	CH / EH	Digital		Usually the ODD channel (n+1)
Turret	**`turretHoodServo`**	Servo	Servo	CH / EH	Servo		Tilts the launch up/down
Camera	**`limelight`**	Limelight 3A	Limelight 3A	**CH**	USB	**USB 3.0**	Must be the USB 3.0 port
FlowerFlipper	**`flowerFlipperServo`**	Servo	Servo	CH / EH	Servo		Shakes back and forth
Lights	**`leftIndicatorLight`**	goBILDA RGB Indicator Light	Servo	CH / EH	Servo		Black wire to ground mark
Lights	**`rightIndicatorLight`**	goBILDA RGB Indicator Light	Servo	CH / EH	Servo		Black wire to ground mark

Port types: Motor = 4 per hub (0-3). **Encoder** = the small 4-pin socket next to each motor port, same number. **Servo** = 6 per hub (0-5). **Digital** = 4 sockets per hub, each with 2 channels (0-1, 2-3, 4-5, 6-7): write the **channel** the config uses. **I2C bus** = 4 per hub (0-3). **Config type** = what to pick in the Driver Station when adding the device.

Reminders: the Expansion Hub connects with the RS485 **3-pin** cable plus XT30 power. The turret encoder goes on an Expansion Hub encoder port whose motor port (same number) is **empty**. Drive motor encoders and the flywheel encoder use their own motor's encoder port. Nothing may move during INIT (G304.H).

Hub Ports

Write the **config name** next to each port as you plug things in, so anyone can trace a wire. Count check: **5 motors** (8 ports), **8 servos** (12 ports), **3 digital** (16 channels), **1 encoder** on its own (the ELC, on an **Expansion Hub** encoder port whose motor port stays empty).

CONTROL HUB (CH)		
Motor (0-3) | right column: its **Encoder** port (same number)		
Motor 0		enc 0:
Motor 1		enc 1:
Motor 2		enc 2:
Motor 3		enc 3:
Servo (0-5)		
Servo 0		
Servo 1		
Servo 2		
Servo 3		
Servo 4		
Servo 5		
Digital (channels 0-7 (ports 0-1, 2-3, 4-5, 6-7))		
Digital 0		
Digital 1		
Digital 2		
Digital 3		
Digital 4		
Digital 5		
Digital 6		
Digital 7		
Analog (0-3)		
Analog 0		
Analog 1		
Analog 2		
Analog 3		
I2C bus (0-3)		
I2C bus 0	**imu (built in)**	
I2C bus 1		
I2C bus 2		
I2C bus 3		
USB		
USB 3.0	**limelight** (must be this one)	
USB 2.0		

EXPANSION HUB (EH)		
Motor (0-3) | right column: its **Encoder** port (same number)		
Motor 0		enc 0:
Motor 1		enc 1:
Motor 2		enc 2:
Motor 3		enc 3:
Servo (0-5)		
Servo 0		
Servo 1		
Servo 2		
Servo 3		
Servo 4		
Servo 5		
Digital (channels 0-7 (ports 0-1, 2-3, 4-5, 6-7))		
Digital 0		
Digital 1		
Digital 2		
Digital 3		
Digital 4		
Digital 5		
Digital 6		
Digital 7		
Analog (0-3)		
Analog 0		
Analog 1		
Analog 2		
Analog 3		
I2C bus (0-3)		
I2C bus 0		
I2C bus 1		
I2C bus 2		
I2C bus 3		